Cambridge Instruments & Engg. Co.

EST. 1963 · HOWRAH, INDIA

INSULATION TESTER · HAND-DRIVEN · BAKELITE BODY

CIE/555

Compact, silent, moulded bakelite body

Direct-reading insulation resistance testing

A compact, portable instrument for direct reading of insulation resistance on electrical operating machines, rotating plant, stationary apparatus, insulation control and switching equipment, distribution networks, communication cables, insulators, transformers, industrial installations, cables, wiring and other devices capable of carrying the rated voltage. It is a versatile instrument for locating intermittent shorts, defective electrical connections, insulation failure and conductor failure, and for preventing costly repairs and breakdowns.



CIE/555 · ILLUSTRATIVE IMAGE

11	4	3 s
STANDARD RANGES	RATED VOLTAGES	RESPONSE TIME

STANDARD RATED TEST VOLTAGES

100 V · 250 V · 500 V · 1,000 V

01 HIGHLIGHTS

- ✓ Moulded high-grade bakelite body
- ✓ Collapsible driven handle, folds back flush into the casing
- ✓ High-sensitivity cross-coil indicator
- ✓ Two push buttons for connection, accepting standard cable or 4 mm plug
- ✓ Accurate and quick response (3 seconds)
- ✓ Silent generator
- ✓ Chuck nut locking arrangement prevents anti-clockwise rotation
- ✓ Rubber shoes for stability on the working surface
- ✓ Graceful appearance
- ✓ Conforms to IS 2992:1987

02 TYPICAL APPLICATIONS

- Insulation testing of electrical operating machines
- Motors, generators, and other rotating machinery
- Insulation control and switching equipment testing
- Distribution network testing
- Communication cable insulation testing
- Insulator and transformer insulation checks
- Industrial installation and wiring checks
- Locating intermittent shorts and defective electrical connections

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03 HOW IT WORKS

PRINCIPLE

A hand-driven clutch-type A.C. generator and a highly sensitive cross-coil indicator. The generated A.C. voltage is converted to D.C. by silicon diodes.

CONSTRUCTION

Moulded in high-grade bakelite: compact, dust-free and economical. Rubber shoes hold the surface and give the instrument stability, and two push-button connections accept a standard cable or a 4 mm plug.

04 SALIENT FEATURES

Silent generator

The A.C. generator runs extremely smoothly and silently thanks to the dynamic balance of its rotating parts, and is free from mechanical noise.

Centrifugal governor

A high-grade centrifugal governor holds the voltage constant for accurate measurement. Variation in crank speed does not affect the reading, and a cut-off clutch disengages the drive beyond the rated speed.

Collapsible handle

The generator handle folds back flush into the casing when the tester is not in use.

Quick response

3 seconds, when a resistance equal to the central scale mark is suddenly inserted between the measurement terminals, as per IS 2992:1987.

05 TECHNICAL SPECIFICATIONS

Body	Moulded high-grade bakelite
Generator Type	Hand-driven, clutch-type generator
Indicator	High-sensitivity cross-coil
Scale Length	75 mm (approx.)
Accuracy	Conforms to IS: 2992-1987
Rotation Speed	160 r.p.m.
Response Time	3 seconds
Weight	1.4 kg (approx.)
Dimensions (L×W×H)	155 × 104 × 70 mm (approx.)
H.V. Test	Withstands minimum 2000 V or (Rated Voltage + 1000) V A.C. (rms) between electrical circuit and handle, for 1 minute
Insulation Resistance Test	Better than 50 MΩ at 500 V between electrical circuit and containing case

Ranges are listed on the following pages. Specifications are approximate where marked.

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06 STANDARD RANGES BY RATED VOLTAGE

RATED VOLTAGE	INSULATION RESISTANCE RANGE (MΩ)
100 V	0–10 · 0–20
250 V	0–20 · 0–50
500 V	0–50 · 0–100 · 0–200 · 0–500
1,000 V	0–100 · 0–200 · 0–500

Ranges shown are the standard builds. Any other range can be manufactured against specific requirements.

07 OPTIONAL ACCESSORIES (AT EXTRA COST)

- Carrying case (bakelite body)
- Testing leads, 1 m

08 HOW TO ORDER

To order, quote the **model**, the **rated voltage** and the **insulation range**, for example `CIE/555, 500 V / 0–500 MΩ`. Any range not listed can be manufactured against specific requirements.

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